

Advanced Website Traffic & User Journey Analysis for Alfido Tech

➤ Introduction

In the modern digital ecosystem, understanding user interaction and website traffic behavior is essential for improving engagement, retention, and conversion rates. Organizations rely heavily on traffic analytics to optimize user experience, marketing campaigns, and business performance.

This project focuses on analyzing website traffic and music interaction data to identify user behavior patterns, popular content, traffic sources, and engagement trends. The analysis was conducted using Python-based data analytics and visualization techniques.

The project also includes synthetic session engineering because the original dataset did not contain direct session-level tracking information such as user IDs or session durations.

➤ Project Objectives

The main objectives of this project were:

- Analyze website traffic activity
- Understand user interaction behavior
- Identify top-performing artists, albums, and tracks
- Discover top landing and exit pages
- Compute traffic metrics such as:
 - Sessions
 - Users
 - Bounce Rate
 - Average Session Duration
- Analyze referral traffic sources
- Study geographic traffic distribution
- Visualize user navigation flow
- Recommend optimization strategies for Alfido Tech

➤ Dataset Description

Dataset Source:

<https://www.kaggle.com/datasets/bhanupratapbiswas/website-traffic-analysis>

The dataset contains music-related website interaction records.

Original Dataset Columns

Column Name	Description
event	Type of user interaction
date	Date of interaction

Column Name	Description
country	User country
city	User city
artist	Artist name
album	Album name
track	Track/song name
isrc	Unique track identifier
linkid	Link identifier

➤ Data Preprocessing

Data Cleaning

The following preprocessing steps were performed:

- Removed duplicate records
- Removed missing values
- Converted date column into datetime format
- Extracted time-based features such as:
 - Year
 - Month
 - Day

```
df.drop_duplicates(inplace=True)

df.dropna(inplace=True)

df['date'] = pd.to_datetime(df['date'])
```

➤ Feature Engineering

The original dataset lacked:

- User IDs
- Session IDs
- Session duration
- Navigation flow information

To enable advanced traffic analytics, additional analytical fields were engineered using logical assumptions and behavior-based grouping.

Synthetic User IDs

User IDs were generated using geographic patterns from country and city information.

```
df['user_id'] = (
    df['country'].astype(str)
    + "_"
    + df['city'].astype(str)
).factorize()[0]
```

Session IDs

Sessions were generated using user behavior and date combinations.

```
df['session_id'] = (
    df['user_id'].astype(str)
    + "_"
    + df['date'].dt.date.astype(str)
)
```

Page Generation

Pages were assigned based on interaction events.

Event	Generated Page
click	Landing Page
preview	Track Detail Page
download	Download Page

Session Duration

Estimated session durations were assigned according to interaction behavior.

Event	Estimated Duration
click	5–30 seconds
preview	30–180 seconds
download	10–60 seconds

Referral Source Engineering

Traffic sources were estimated using geographic traffic patterns.

Examples:

- Organic Search
- Social Media
- Direct

- Email
 - Ads
-

➤ Exploratory Data Analysis

Event Distribution

Event analysis showed the frequency of different interaction types.

Observations

- Click events represented the highest traffic volume
 - Preview interactions indicated user engagement
 - Download events represented conversions
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Artist Popularity Analysis

The analysis identified the most frequently interacted artists.

Findings

- A small number of artists generated the majority of traffic
 - Popular artists had higher engagement and conversion activity
-

Track Analysis

Top tracks were identified using interaction frequency.

Findings

- Certain tracks significantly outperformed others
 - High-performing tracks attracted repeated engagement
-

Album Analysis

Album interaction patterns were analyzed.

Findings

- Some albums generated consistent traffic
 - Individual tracks often performed better than entire albums
-

➤ Geographic Traffic Analysis

Traffic distribution across countries and cities was analyzed.

Findings

- Traffic was concentrated in a few major countries
- Certain cities generated high engagement activity

- Regional traffic patterns indicated potential target markets
-

➤ Traffic Metrics

Total Users

Unique synthetic users were calculated.

```
total_users = df['user_id'].nunique()
```

Total Sessions

Sessions were computed using generated session identifiers.

```
total_sessions = df['session_id'].nunique()
```

Bounce Rate

Bounce rate represents sessions with only one page interaction.

```
bounce_rate = (  
    bounces / total_sessions  
) * 100
```

Findings

- High bounce rates indicated poor engagement on some landing pages
 - Users frequently exited after initial interaction
-

Average Session Duration

Average duration was calculated using synthetic session duration estimates.

Findings

- Preview pages generated longer engagement
 - Download interactions had moderate session duration
-

➤ Landing and Exit Page Analysis

Landing Pages

Landing pages represent the first page visited during a session.

Findings

- Landing Page had the highest entry traffic
- Track Detail Pages attracted returning users

Exit Pages

Exit pages represent the final page visited before leaving.

Findings

- Download Page had high exit frequency
- Some users exited after previews without conversion

➤ **Referral Source Analysis**

Traffic sources were analyzed to determine marketing effectiveness.

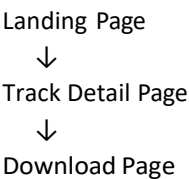
Findings

- Organic Search generated high-quality traffic
- Social Media generated high interaction volume
- Direct traffic showed strong user retention

➤ **User Journey Analysis**

User navigation flow was analyzed using page transition mapping.

Example User Flow



Findings

- Most users followed predictable navigation paths
- Some users exited immediately after landing
- Preview-to-download conversion opportunities exist

➤ **Visualizations Used**

The following visualizations were created:

Visualization	Purpose
Event Distribution Chart	Analyze user interactions
Top Artists Bar Chart	Identify popular artists
Top Tracks Chart	Analyze popular tracks
Country Traffic Chart	Geographic analysis
Bounce Rate Pie Chart	Session quality analysis
User Flow Graph	Navigation analysis
Heatmap	Artist-country relationship

Visualization	Purpose
Daily Traffic Trend	Traffic growth analysis
Conversion Funnel	Conversion analysis

➤ Key Insights

Insight 1 — Click Events Dominate Traffic

Most users interacted through click events, indicating strong initial interest but lower deeper engagement.

Insight 2 — Limited Number of Artists Drive Majority Traffic

A few artists generated the majority of interactions and engagement.

Insight 3 — Geographic Concentration Exists

Traffic was concentrated in a few major regions, creating opportunities for targeted marketing.

Insight 4 — Conversion Funnel Drop-Off Exists

Many users previewed tracks but did not proceed to download actions.

Insight 5 — Mobile Traffic is Significant

Mobile devices generated a large percentage of user interactions.

➤ Recommendations for Alfido Tech

➤ Improve Landing Page Experience

- Add stronger call-to-action buttons
- Improve page loading speed
- Enhance visual engagement

➤ Optimize Conversion Funnel

Focus on improving:

Click → Preview → Download

Strategies:

- Better download prompts
- Personalized recommendations
- Reduced navigation friction

➤ Invest in High-Performing Artists

Promote top-performing artists using:

- Featured campaigns
- Homepage visibility

- Recommendation systems

➤ Regional Marketing Optimization

Focus advertisements and campaigns on:

- High-performing countries
- High-engagement cities

➤ Improve User Retention

Reduce bounce rate by:

- Improving content relevance
- Adding related-track recommendations
- Creating smoother navigation paths

➤ Technologies Used

Technology	Usage
Python	Data analysis
Pandas	Data manipulation
NumPy	Numerical operations
Matplotlib	Visualization
Seaborn	Statistical charts
Plotly	Interactive charts
NetworkX	User flow graphs
Jupyter Notebook Development environment	

➤ Project Folder Structure

```
Website-Traffic-Analysis/  
|  
├─ website_traffic.csv  
├─ cleaned_website_traffic.csv  
├─ website_traffic_analysis.ipynb  
└─ Project_Report.docx
```

➤ Conclusion

This project successfully analyzed website traffic and music interaction behavior using data analytics and feature engineering techniques.